

Calculate the Laplacian of the following functions:

(a) $T_a = x^2 + 2xy + 3z + 4$

(b) $T_b = \sin x \sin y \sin z$

(c) $T_c = e^{-5x} \sin 4y \cos 3z$

(d) $\mathbf{v} = x^2 \hat{\mathbf{x}} + 3xz^2 \hat{\mathbf{y}} - 2xz \hat{\mathbf{z}}$

Solution:

The Laplacian of a function T is defined as

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}$$

and the Laplacian of a vector $\mathbf{v}$ is defined as

$$\nabla^2 \mathbf{v} = (\nabla^2 v_x) \hat{\mathbf{x}} + (\nabla^2 v_y) \hat{\mathbf{y}} + (\nabla^2 v_z) \hat{\mathbf{z}}$$

Using this, the result is

(a) $\nabla^2 T_a = 2$

(b) $\nabla^2 T_b = -3 \sin x \sin y \sin z$

(c) $\nabla^2 T_c = e^{-5x} \sin 4y \cos 3z (25 - 16 - 9) = 0$

(d) $\nabla^2 \mathbf{v} = 2\hat{\mathbf{x}} + 6x\hat{\mathbf{y}}$

□
